

Technology Stack

Date	6 July 2026
Team ID	SWTID-2026-8977
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client Side (e.g., React, Angular, HTML/CSS)	HTML5, CSS3, JavaScript	Provides a responsive and user-friendly interface for students to ask questions, generate quizzes, summarize content, and view personalized learning paths.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, FastAPI	Handles user requests, processes inputs, communicates with the Google Gemini API, and returns AI-generated educational responses efficiently with high performance.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	SQLite	Stores user queries, learning history, quiz results, and personalized learning progress securely for future reference.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Google AI Studio (Gemini API), Local Deployment / Render / Railway	Provides access to Google Gemini AI services and supports deployment of the FastAPI application for reliable and scalable usage.

5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	Git, GitHub	Manages source code, tracks project versions, and enables efficient development and deployment.
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Google Gemini API, FastAPI, Uvicorn, Python-dotenv	Google Gemini API generates AI-powered educational content; Uvicorn runs the FastAPI server; Python-dotenv securely manages API keys and environment variable.